

The HIV-derived transactivator of transcription (Tat) is a short peptide with the sequence RKKRRQRRPPQ. When paired with liposomes (ie, synthetic vesicles) composed of cationic lipids, Tat can transport large molecules across cell membranes and has been studied as a vector for gene therapy. However, Tat delivers cargo to cells with relatively low efficiency. Researchers hypothesized that a Tat homodimer could yield stronger DNA binding than the monomer, which may improve the efficiency of DNA delivery.

To study this hypothesis, Tat and two variant peptides (Variants 1 and 2) were produced by solid-phase peptide synthesis, in which the growing peptide chain is immobilized on polystyrene beads. Unlike biological systems, solid-phase synthesis assembles peptides from the C-terminus to the N-terminus, as shown in Figure 1. Therefore, in the case of Tat, the first amino acid added to the chain is glutamine, followed by proline, and so on.

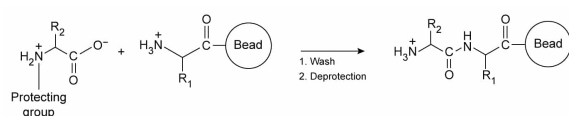


Figure 1 Simplified schematic of one round of solid-phase peptide synthesis. The protecting group is removed during deprotection. The process is repeated until the peptide is complete.

Variants 1 and 2 contained a C-terminal and an N-terminal cysteine residue, respectively. Each variant was allowed to dimerize by disulfide bond formation. Peptide-DNA binding interactions were then studied by mixing varying amounts of each peptide with plasmid DNA encoding luciferase protein (pLuc). The resulting peptide-DNA complexes were run on agarose gels, with results displayed in Figure 2.

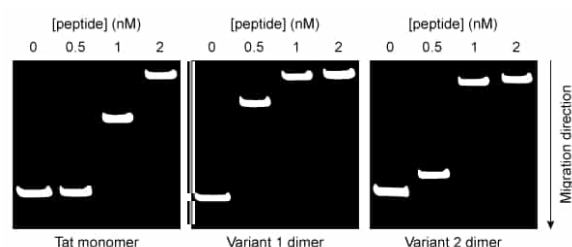


Figure 2 Agarose gel results for pLuc plasmid exposed to varying amounts of each peptide

Equal concentrations of each type of peptide-DNA complex were then incubated with cationic liposomes and exposed to cultured cells for 24 hours. The cells were then assessed for luciferase activity. Greater luciferase activity indicated more efficient delivery of the pLuc plasmid. The results are shown in Figure 3.

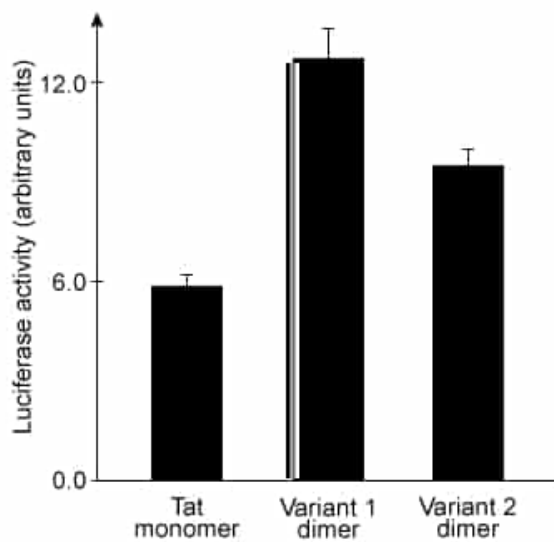


Figure 3 Plasmid delivery efficiency of three peptides, as measured by induced luciferase activity

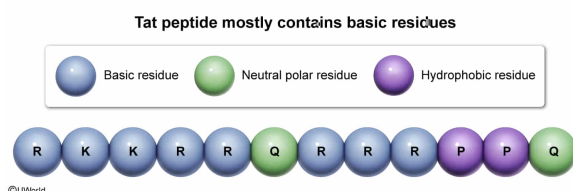
The Tat peptide consists primarily of amino acids with side chains that are:

- ☐ A. Aromatic [1%]
- ☐ B. Hydrophobic [6%]
- ☐ C. Acidic [8%]
- ✓ ☒ D. Basic [84%]

Correct

83% answered correctly

Explanation:



Amino acids have identical **backbone structures** and are distinguished by the structures of their side chains (ie, R groups). The side chains are broadly classified as either **hydrophobic** or **hydrophilic**. Hydrophilic amino acids may be further categorized as **acidic** (negatively charged at physiological pH), **basic** (positively charged at physiological pH), or **neutral polar**. Hydrophobic amino acids are often classified as **aliphatic** or **aromatic**.

The sequence of Tat given in the passage shows that 8 of the 12 amino acids in the peptide are either **arginine** (R) or **lysine** (K), with a few **proline** (P) and **glutamine** (Q) residues present. Both **R and K residues are basic** (ie, they accept protons and carry a positive charge at physiological pH). Therefore, the Tat peptide is primarily composed of amino acids with **basic side chains**.

(Choice A) The aromatic amino acids are **tryptophan** (W), **phenylalanine** (F), and **tyrosine** (Y). The Tat peptide does not contain any aromatic residues.

(Choice B) The only hydrophobic amino acid in the Tat peptide is proline (P), which contributes only 2 of the 12 residues in the peptide.

(Choice C) **Glutamic acid (ie, protonated glutamate)** (E) and **aspartic acid (ie, protonated aspartate)** (D) are the only amino acids considered to have acidic side chains. Neither is present in the Tat peptide.

Educational objective:

Amino acids are classified broadly as hydrophobic or hydrophilic. They may be further classified as acidic, basic, or neutral (for hydrophilic amino acids), or as aliphatic or aromatic (for hydrophobic amino acids).

Time spent:0 sec

Subject:
Biochemistry

QID:403066

System:
Amino Acids and Proteins

The HIV-derived transactivator of transcription (Tat) is a short peptide with the sequence RKKRRQRRRPPQ. When paired with liposomes (ie, synthetic vesicles) composed of cationic lipids, Tat can transport large molecules across cell membranes and has been studied as a vector for gene therapy. However, Tat delivers cargo to cells with relatively low efficiency. Researchers hypothesized that a Tat homodimer could yield stronger DNA binding than the monomer, which may improve the efficiency of DNA delivery.

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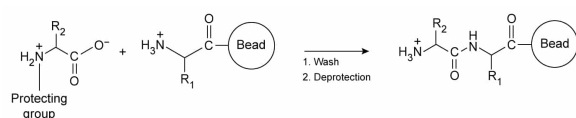


Figure 1 Simplified schematic of one round of solid-phase peptide synthesis. The protecting group is removed during deprotection. The process is repeated until the peptide is complete.

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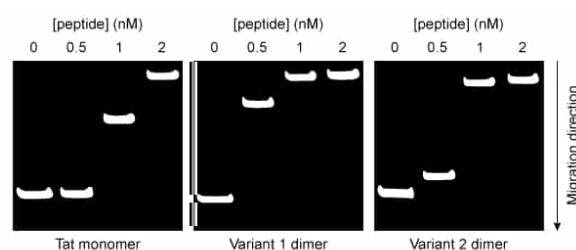


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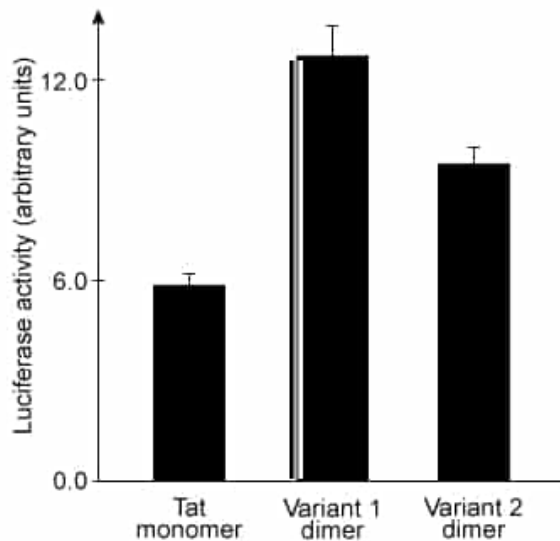


Figure 3 Plasmid delivery efficiency of three peptides, as measured by induced luciferase activity

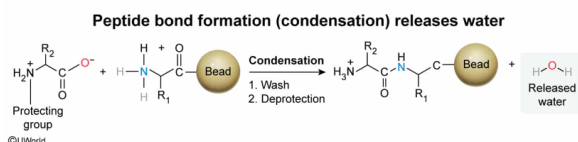
Which of the following best describes the reaction between amino acids during solid-phase peptide synthesis?

- ☒ A. It is a condensation reaction, in which water is released [81%]
- ☐ B. It is a condensation reaction, in which water is consumed [7%]
- ☐ C. It is a hydrolysis reaction, in which water is released [7%]
- ☐ D. It is a hydrolysis reaction, in which water is consumed [2%]

Correct

82% answered correctly

Explanation:



Peptide bonds are **amide** bonds that form between the N-terminus of one amino acid or peptide and the C-terminus of another. At physiological pH, the **N- and C-termini** consist of an NH_3^+ group and a COO^- group, respectively. When these groups react with each other, they **condense** to form an amide group with the formula NHCO . Therefore, 1 O atom and 2 H atoms are lost from the reacting species. These atoms combine to **form water**.

Solid-phase peptide synthesis such as that shown in Figure 1 is an example of peptide bond formation. As such, this synthesis is a **condensation reaction** in which **water is released**.

(Choice B) Condensation reactions release water. They do not consume water.

(Choices C and D) **Hydrolysis** *breaks* peptide bonds by adding water across them. Solid-phase peptide synthesis results in *forming*, not breaking, peptide bonds.

Educational objective:

Peptide bonds form by condensation reactions in which water is released. The reverse reaction, hydrolysis, consumes water to break the peptide bond.

Time spent:0 sec

Subject:
Biochemistry

QID:403067

System:
Amino Acids and Proteins

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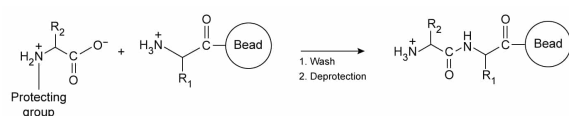


Figure 1 Simplified schematic of one round of solid-phase peptide synthesis. The protecting group is removed during deprotection. The process is repeated until the peptide is complete.

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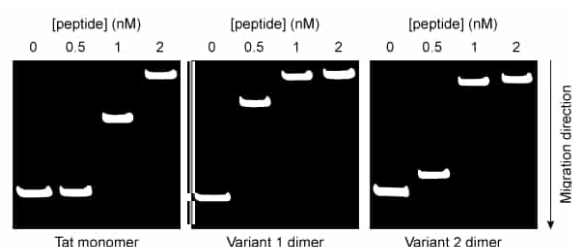


Figure 2 Agarose gel results for pLuc plasmid exposed to varying amounts of each peptide

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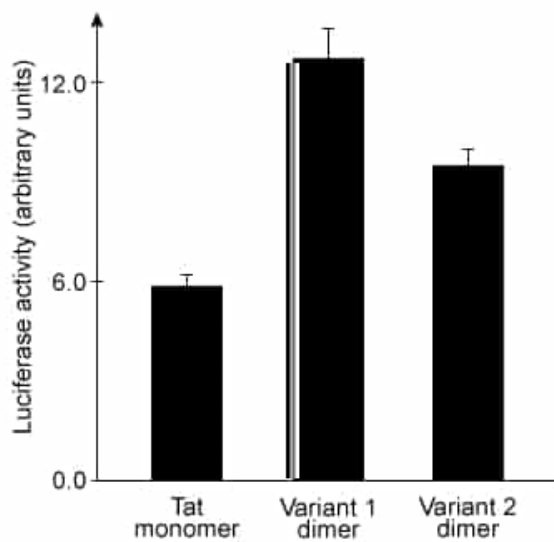


Figure 3 Plasmid delivery efficiency of three peptides, as measured by induced luciferase activity

The researchers observed that the last three amino acids added to the Tat monomer during solid-phase peptide synthesis required more time for linkage than the first several amino acids added. Which of the following best explains this observation?

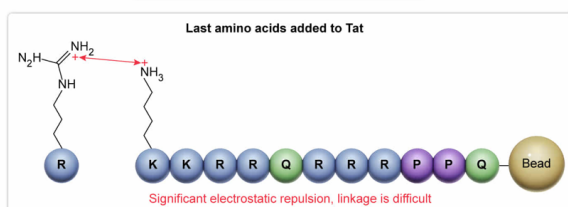
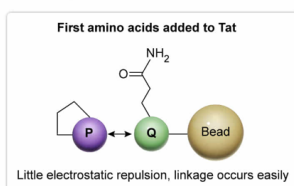
- ☐ A. The amino groups of the last three amino acids are less able to react with the carboxyl group of the growing peptide. [8%]
- ☒ B. The side chains of the three N-terminal amino acids experience stronger electrostatic repulsions than the side chains of the three C-terminal amino acids. [54%]
- ☐ C. The last three amino acids added to the peptide include the proline residues near the C-terminus, which add kinks to the chain. [32%]
- ☐ D. Amino acids are generally less likely to collide with longer peptides than they are with shorter peptides. [4%]

Correct

54% answered correctly

Explanation:

Electrostatic repulsions between amino acid side chains



The side chains of the amino acids in a peptide **interact with each other** through **electrostatic interactions**, including **ionic attractions and repulsions**, **hydrogen bonding**, **London dispersion forces**, and **steric interference**. These interactions can also influence the ability of amino acids and peptides to react with each other. Amino acids with **similar charges repel** each other, reducing the probability that they will collide in a way that results in a reaction (ie, a **productive collision**).

The passage states that **solid-phase peptide synthesis occurs from C-terminus to N-terminus**. However, by convention the sequence of a peptide or protein is always *written from N-terminus on the left to C-terminus on the right*. Therefore, the last three amino acids added to the Tat monomer (ie, those at the N-terminus) were a **lysine** (K), another lysine, and an **arginine** (R) residue. These residues are all positively charged and will repel each other, resulting in fewer productive collisions.

In contrast, the first several amino acids added (at the C-terminus) were **glutamine** (Q), **proline** (P), and another proline, which are neutral and will not experience the same electrostatic repulsion. These amino acids are more likely to collide productively and react more quickly.

(Choice A) Figure 1 shows that it is the amino group of the *growing peptide* (not the added amino acids) that reacts, and that the *carboxyl group* (not the amino group) of added amino acids reacts.

(Choice C) Proline residues are known to add kinks to polypeptide chains, making them unfavorable for **α -helices**. However, because the proline residues in Tat are near the C-terminus of the peptide, they are among the *first* amino acids added, not the last.

(Choice D) Although charge repulsions make collisions less likely at the end of the Tat peptide, this decreased probability is *not* related to the *length* of the growing chain. In general, the length of a peptide chain has no impact on the probability of collisions between molecules.

Educational objective:

Charged amino acids within a peptide interact with each other. Similarly charged residues repel each other whereas oppositely charged residues attract each other.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403068

System:
Amino Acids and Proteins

The HIV-derived transactivator of transcription (Tat) is a short peptide with the sequence RKKRRQRRRPPQ. When paired with liposomes (ie, synthetic vesicles) composed of cationic lipids, Tat can transport large molecules across cell membranes and has been studied as a vector for gene therapy. However, Tat delivers cargo to cells with relatively low efficiency. Researchers hypothesized that a Tat homodimer could yield stronger DNA binding than the monomer, which may improve the efficiency of DNA delivery.

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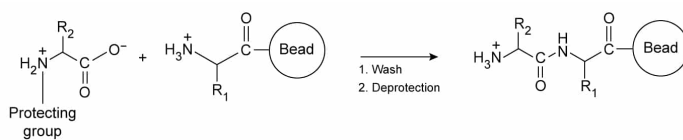


Figure 1 Simplified schematic of one round of solid-phase peptide synthesis. The protecting group is removed during deprotection. The process is repeated until the peptide is complete.

Variants 1 and 2 contained a C-terminal and an N-terminal cysteine residue, respectively. Each variant was allowed to dimerize by disulfide bond formation. Peptide-DNA binding interactions were then studied by mixing varying amounts of each peptide with plasmid DNA encoding luciferase protein (pLuc). The resulting peptide-DNA complexes were run on agarose gels, with results displayed in Figure 2.

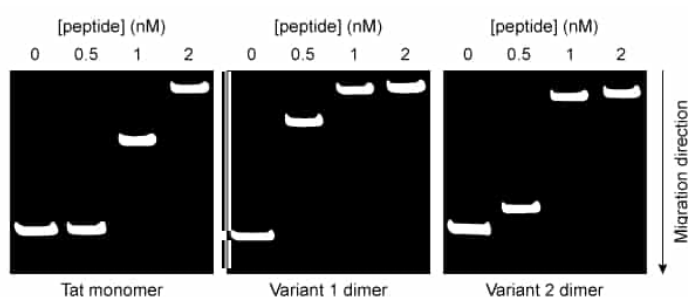


Figure 2 Agarose gel results for pLuc plasmid exposed to varying amounts of each peptide

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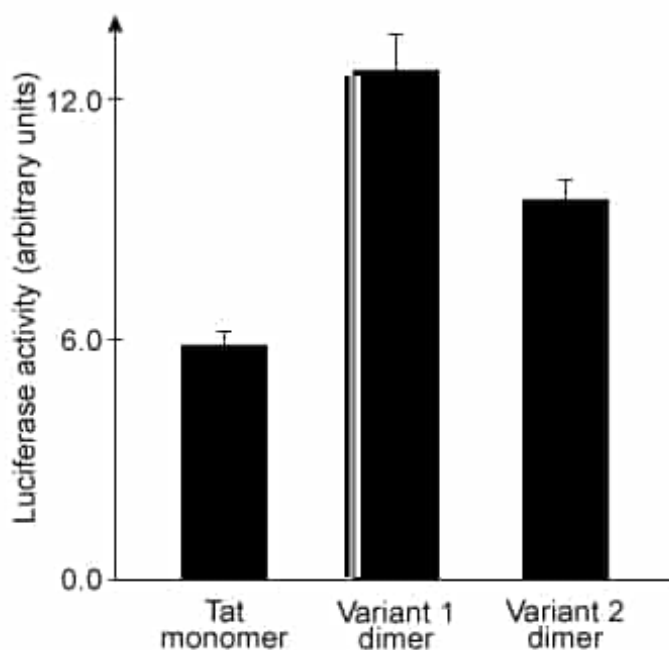


Figure 3 Plasmid delivery efficiency of three peptides, as measured by induced luciferase activity

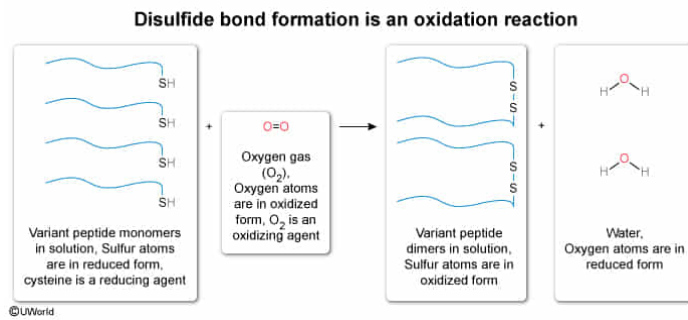
The addition of which reagent would result in the greatest amount of dimerization of the Tat variants?

- ☐ A. A strong acid such as HCl [10%]
- ☐ B. A strong base such as NaOH [15%]
- ☐ C. A reducing agent such as dithiothreitol [22%]
- ✓ ☒ D. An oxidizing agent such as oxygen gas [50%]

Correct

50% answered correctly

Explanation:



Cysteine (C) residues in proteins and peptides can react with each other to form **disulfide bonds**. These bonds are commonly (although not always) present in **dimers and multimers** and help hold protein subunits together. The formation of a disulfide bond occurs through an **oxidation-reduction reaction**, in which the sulfur atoms of cysteine residues are initially in reduced form and become oxidized. For this reaction to occur, an **oxidizing agent must be present**.

The passage indicates that the Tat variants dimerize by disulfide bond formation. Therefore, the addition of an **oxidizing agent such as oxygen gas** would help **increase the amount of disulfide bond formation** and, as a result, the amount of **dimerization that occurs**.

(Choices A and B) The acidity or basicity of a chemical is not directly related to that chemical's ability to act as an oxidizing agent. Disulfide bond formation requires an oxidizing agent, not an acid or a base.

(Choice C) Reducing agents such as dithiothreitol are commonly **used in gels** to *break* disulfide bonds, not form them.

Educational objective:

The formation of disulfide bonds between cysteine residues is an oxidation event and requires the presence of an oxidizing agent, such as oxygen gas, to occur.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403069

System:
Amino Acids and Proteins

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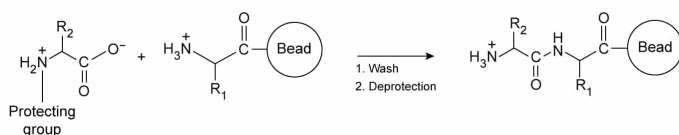


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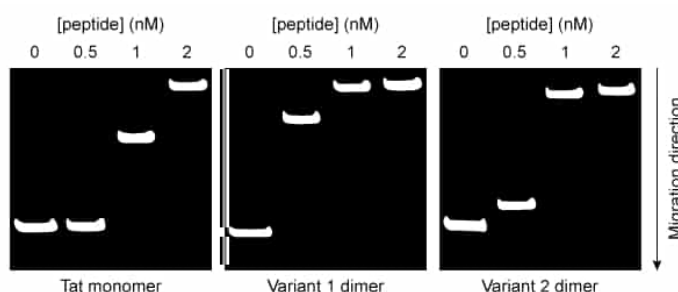


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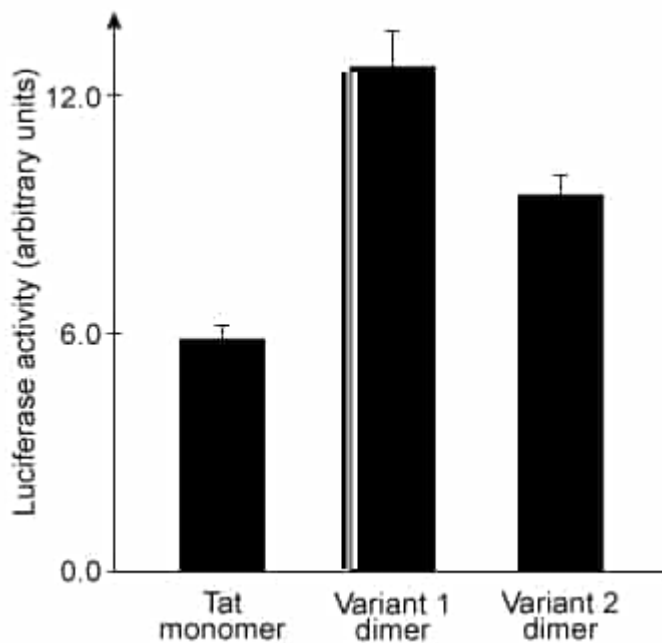


Figure 3 Plasmid delivery efficiency of three peptides, as measured by induced luciferase activity

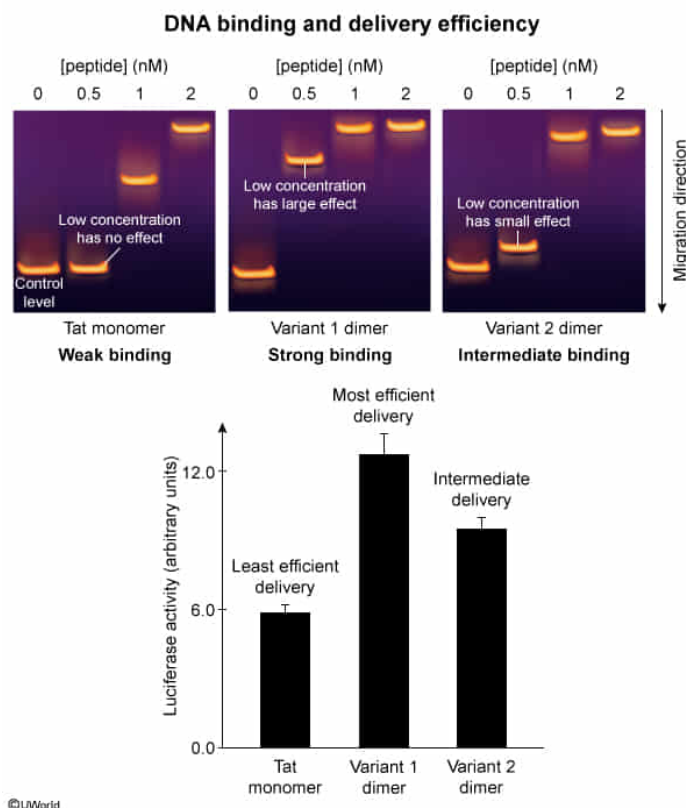
Do the data in Figure 2 and Figure 3 support the hypothesis that stronger peptide-DNA interactions yield more efficient plasmid delivery?

- ✔ ☒ A. Yes; the order of binding strength shown in Figure 2 is the same as the order of delivery efficiency shown in Figure 3. [56%]
- ☐ B. No; the Variant 2 dimer shows the weakest DNA binding but is more efficient than the Tat monomer at cargo delivery. [8%]
- ☐ C. No; the Tat monomer shows the strongest DNA binding but the least efficient plasmid delivery. [16%]
- ☐ D. No; the Variant 2 dimer binds DNA with greater strength than the Variant 1 dimer, but Variant 1 has the most efficient plasmid delivery. [19%]

Correct

56% answered correctly

Explanation:



DNA molecules have a net **negative charge** due to the phosphate groups in the **DNA backbone**. This negative charge causes DNA to migrate **toward the anode** of an **electrophoretic cell** through an agarose gel. **Neutralizing** some of the **negative charges** by binding to positively charged amino acid residues **decreases the mobility** of the DNA through the gel. Peptide binding also increases the mass of the complex, further slowing its migration. As more charges are neutralized and more mass is

creasing the migration. As more charges are neutralized and more mass is added, the DNA migrates progressively **shorter distances** through the gel.

Figure 2 shows that all three forms of the Tat peptide (which are positively charged) cause a decrease in gel mobility. However, different amounts of each form are required to achieve an effect, and the relative amounts needed indicate **binding strength**. The Variant 1 dimer shows a significant decrease in DNA mobility when 0.5 nM peptide is added. The Variant 2 dimer shows a small mobility shift at 0.5 nM but requires 1 nM peptide for a significant effect. The Tat monomer shows no effect at 0.5 nM.

Therefore, Variant 1 shows the strongest binding, followed by Variant 2, and then the Tat monomer.

Figure 3 shows that Variant 1 yields the most efficient DNA delivery, followed by Variant 2, and then the Tat monomer. Therefore, the order of binding strength is the same as the order of delivery efficiency. This supports the hypothesis that stronger peptide-DNA binding is associated with increased delivery efficiency by Tat peptides.

(Choice B) Variant 2 shows *stronger* (not weaker) binding than the Tat monomer.

(Choice C) The Tat monomer shows the *weakest* (not strongest) binding.

(Choice D) Variant 2 binds DNA with *less* (not greater) strength than Variant 1.

Educational objective:

The negative charge of DNA causes it to migrate toward the anode of an electrophoretic cell. Neutralizing the negative charge with positively charged residues and increasing the size of the complex decreases the DNA's gel mobility.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403070

System:
Amino Acids and Proteins

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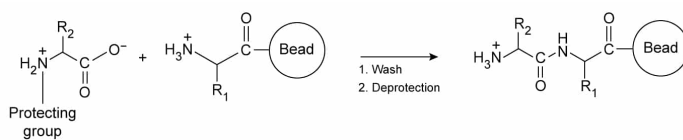


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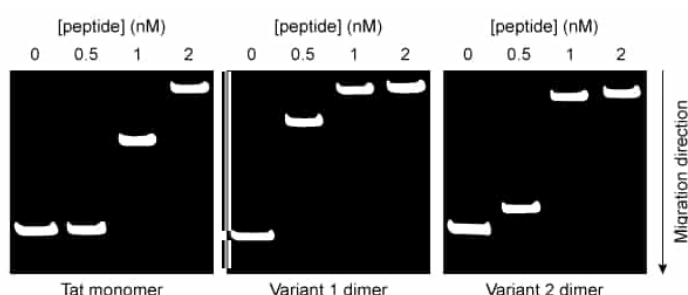


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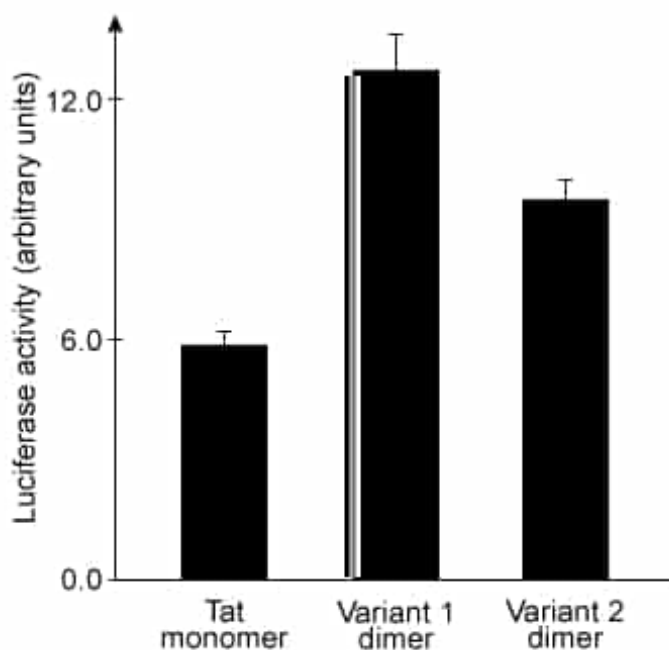


Figure 3 Plasmid delivery efficiency of three peptides, as measured by induced luciferase activity

A researcher hypothesized that cargo delivery is most efficient when the peptide-DNA complex forms prior to DNA interactions with liposomes. Which additional observation would best support this hypothesis?

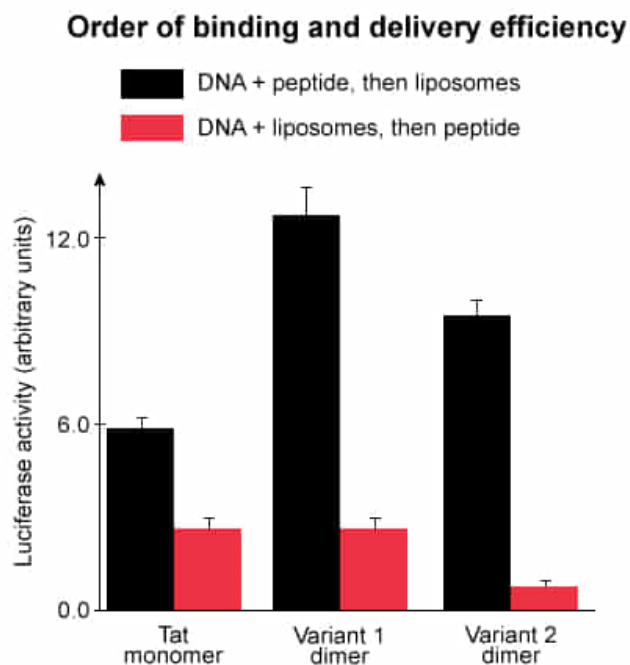
- ☐ A. An increase in luciferase activity when the concentration of liposomes is increased. [5%]
- ☐ B. A decrease in luciferase activity when no peptide is included in the experiment. [8%]
- ☐ C. An increase in luciferase activity when the ratio of liposomes to DNA molecules is increased. [10%]
- ☒ D. A decrease in luciferase activity when liposomes are mixed with

DNA prior to addition of peptides. [74%]

Correct

74% answered correctly

Explanation:



Decreased luciferase activity would indicate that efficiency is decreased when DNA is not allowed to bind Tat variants prior to interactions with liposomes

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Biological complexes consisting of multiple molecules often **must assemble in a particular order**. In the case of Tat, DNA, and liposomes, a researcher hypothesized that Tat or its variants must bind to DNA *before* DNA interacts with cationic liposomes to yield optimal DNA delivery. If this hypothesis is correct, then DNA binding to cationic

liposomes *prior* to interactions with peptides would result in *decreased* DNA delivery.

Figure 3 was generated by mixing existing peptide-DNA complexes (ie, the peptide and DNA are already interacting) with liposomes and then incubating with cells. Mixing DNA with cationic liposomes *prior* to addition of peptides would allow DNA-liposome interactions to occur *before* DNA can interact with peptides. If such interactions interfere with DNA delivery (as the researcher suggests), then less luciferase will be produced and less activity will be observed. This observation would provide strong support for the researcher's hypothesis.

(Choices A and C) Increasing the concentration of liposomes (or the ratio of liposome to DNA) would not change the fact that the DNA and peptide are already interacting (ie, forming a complex) prior to liposome addition, so these experiments would not test the hypothesis.

(Choice B) Excluding peptide from the assay to establish baseline levels of DNA delivery is an important control but does not test the importance of the *order* in which DNA must interact with peptides and liposomes.

Educational objective:

The order of interactions between biological molecules can affect the outcome of an experiment. This effect can be monitored by changing the order in which each component is added to an experimental system.

Time spent:0 sec

Subject:
Biochemistry

QID:403071

System:
Amino Acids and Proteins